An Entropy-Weighted Network for Polar Sea Ice Open Lead Detection From Sentinel-1 SAR Images

Zeyu Liang, Xiaoping Pang, Qing Ji, Xi Zhao, Guoyuan Li, and Yizhuo Chen

***Abstract*— Sea ice leads in the Arctic Ocean and Antarctic Ocean are of great significance to polar ecology, climate change, and ship navigation. While rule-based remote-sensing classification methods, such as the threshold method, have been widely used for surveying and mapping polar sea ice leads, they have difficulty in overcoming the problems of noise and poor generalization. In this article, we presented an automated, deep learning sea ice open lead detection network in low wind speed conditions, entropy-weighted network (EW-Net). In EW-Net, a dense block was introduced into a U-Net baseline network to strengthen feature propagation, and an entropy sampling structure was designed to improve the pertinence of the network to leads and alleviate the influence of noise. Furthermore, an entropy-weighted feature fusion block was designed to better restore features. EW-Net was trained on the Sentinel-1 synthetic aperture radar (SAR) dataset in order to obtain lead maps of different polar regions from Sentinel-1 images. The results showed that the proposed EW-Net model was more effective and more generalizable for polar sea ice lead detection than the threshold method and the deep learning methods U-Net and DeepLab V3 Plus, with an overall accuracy (OA) exceeding 0.97. In addition, EW-Net showed advantages in terms of outstanding generalization from cross-sensor image monitoring of leads. The method proposed in this article could be beneficial for providing accurate maps of polar sea ice leads in open water and contribute to further research on polar science.**

***Index Terms*— Antarctic Ocean, Arctic Ocean, deep learning, entropy, sea ice lead, synthetic aperture radar (SAR).**

I. Introduction

A sea ice lead is a linear open water area within pack ice [1]. The World Meteorological Organization (WMO) defines a lead as a crack in ice that is more than 50 m wide and from several kilometers to hundreds of kilometers in length. Sea ice is not homogeneous and it drifts, deforms, and fractures as the result of external forces (wind and ocean currents), which is the main reason for the formation of leads [2]. The opening of leads breaks the continuity of insulating ice and results in stronger air–ocean interaction, which leads to ice production, brine rejection, and heat transfer. During the cold season, thin ice forms quickly in newly opened leads due to the large temperature difference between ocean and atmosphere [3], the strong exchange state tends to be balanced, and the sea surface is recovered by new ice [4]. In the central Arctic, leads only cover 1%–2% of the ocean during winter but account for more than 70% of the upward heat fluxes [2], [5]. The formation and evolution of sea ice leads affect the physical properties of sea ice, sea water and marine ecosystems, and the safe navigation of ships in polar regions, thus engendering scientific research significance [6]–[8].

Satellite remote sensing has become an increasingly important technology for research on polar sea ice leads due to the limitations of polar *in situ* observations [9]. Synthetic aperture radar (SAR) images are the most suitable data for polar sea ice lead observations because of their large-scale and high-spatial-resolution properties under almost all weather conditions [10]. Spatial information of ice and water, such as the relative position of leads in sea ice, the boundary between leads and sea ice, and the direction and texture of leads, can be observed from SAR images [11], [12]. All of this information is mutually influential and plays a crucial role in identifying leads more accurately and effectively. However, currently, sea ice lead information needs to be obtained from SAR images by manual visual interpretation due to the lack of robust automated image interpretation techniques [13].

In recent studies, texture features calculated from SAR images were used to enhance sensitivity and detect sea ice leads. One of the most frequently used features is the gray-level co-occurrence matrix (GLCM) [14], which was proven to be effective for SAR image classification [15], [16]. In addition, external features, such as the copolarization and cross-polarization ratios [17] and the incidence angle [18], were applied to the sea ice classification of SAR images. The commonly used method for remote-sensing image segmentation is rule-based remote-sensing classification methods, such as the threshold method [10]; however, these methods are not applicable to SAR image classification, because in SAR sensor image processing, adjacent pixels' gray levels change randomly in radar echo signals due to coherence. As a result, coherent speckle noise appears and considerably affects the classification of sea ice [19]–[21]. Moreover, rule-based classification methods are limited by the fact that specific rules must be developed for each image, which makes automation difficult to achieve.

Manuscript received April 3, 2021; revised September 28, 2021 and February 12, 2022; accepted April 18, 2022. Date of publication April 22, 2022; date of current version May 17, 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 42076235 and Grant 41876223 and in part by the Special Fund for High Resolution Images Surveying and Mapping Application System under Grant 42-Y30B04-9001-19/21. *(Corresponding authors: Xiaoping Pang; Qing Ji.)*

Zeyu Liang, Xiaoping Pang, Qing Ji, and Yizhuo Chen are with the Chinese Antarctic Center of Surveying and Mapping, Wuhan University, Wuhan 430079, China (e-mail: pxp@whu.edu.cn; jiqing@whu.edu.cn).

Xi Zhao is with the School of Geospatial Engineering and Science, Sun Yat-sen University, Zhuhai 519082, China.

Guoyuan Li is with the Land Satellite Remote Sensing Application Center, Ministry of Natural Resources, Beijing 100048, China.

Digital Object Identifier 10.1109/TGRS.2022.3169892

1558-0644 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:58:59 UTC from IEEE Xplore. Restrictions apply.

TABLE I
PARAMETERS OF THE IMAGES USED WITH THE CORRESPONDING WIND AND TEMPERATURE

Num	Date	Beam Mode	Location	Wind Speed (m/s)	Wind Direction	2m Temprature (K)	Used for
1	Jun. 04, 2019	EW	Training Image in Fig. 1(a)	3.5435	Southeast	255.4784	Training
2	Jul. 10, 2019	EW	Test Image 1 in Fig. 1(a)	4.0520	Northeast	256.5270	
3	May. 30, 2020	EW	Test Image 2 in Fig. 1(b)	6.6732	Southeast	253.7150	
4	Feb. 06,2021	IW	Test Image 3 in Fig. 1(c)	2.5385	Northeast	267.2623	Testing
5	Nov. 03, 2020	EW	Test Image 4 in Fig. 1(c)	1.7240	Northeast	267.6597	
6	Mar. 31, 2020	EW	Test Image 5 in Fig. 1(d)	5.5677	Northeast	256.3967	

Due to the importance of polar sea ice lead information and the advantage of SAR technology for polar regions, automation and high-precision sea ice lead detection methods based on deep learning are of particular interest. As a trending technology in recent years, deep learning has a strong feature self-learning ability, a high degree of freedom for input data, and has been widely used in many scientific fields [22]. Regarding the classification of sea ice and sea water, previous studies used deep learning to classify images at the pixel level without considering the spatial information of ice and water [23], [24]. A fully convolutional neural network (FCN) is a kind of deep learning neural network specifically designed for images. An FCN can extract image information through a 2-D convolution kernel and, similar to human visual perception, considers spatial information. U-Net, which was first applied to medical image segmentation, is a classic application of an FCN in the field of image segmentation [25] that was improved and introduced for remote-sensing images to detect information [26], [27]. DeepLab is a type of FCN developed and constantly updated by Google [28] that has shown good performance in the field of image segmentation [29], [30]. In polar regions, the rapid change in sea ice conditions has an important impact on scientific research and ship navigation. Compared with traditional image classification methods, such as the threshold method, the FCN method can consider more image information and is less affected by human subjectivity, making it more suitable for ice condition monitoring based on remote-sensing images.

In this study, we explore how to precisely and automatically obtain polar sea ice lead information based on deep learning by fully considering the spatial information of features from SAR images. In this article, Section II describes the data used in this article and the proposed sea ice lead detection method and the experimental parameters and precision evaluation indices. The results and a discussion of the experiments are presented in Sections III and IV, respectively. Finally, we conclude our work in Section V.

II. DATA AND METHODS

A. Sentinel-1 Images

The data used in this article to detect polar sea ice leads are derived from SAR images from the Sentinel-1 satellite. Sentinel-1 operates in a near-polar sun-synchronous orbit with an orbital altitude of 693 km and an inclination of 98.18°. This satellite works in the C-band and exhibits better performance for discriminating sea ice and sea water with high spatial resolution. Sentinel-1 SAR images can be downloaded from the website of the Alaska Satellite Facility (https://search.asf.alaska.edu/).

Six Sentinel-1 SAR images were used in this study. The image information is shown in Table I and the image locations are shown in Fig. 1. Wind and temperature parameters were obtained from ERA5 hourly data on single levels from 1979 to present [31].

B. Generation of the Training Dataset for Sea Ice Lead Detection

C-band microwaves are a preferred band for discriminating sea ice from open water [32], [33]. Sentinel-1 SAR images with C-band detection capability have the advantage of high spatial resolution and are available online at no cost, so we generated a sea ice lead training dataset using one Sentinel-1 SAR HH polarization image, which only transformed the image format from the original SAR image.

The sea ice lead training dataset consisted of SAR subimages, lead binary subimages, and entropy subimages. Each SAR subimage was randomly cropped from the SAR image by a slide window with a size of 512 × 512, and 20% of these subimages were subjected to a flipping algorithm to increase the number of lead features for further consideration. Lead binary subimages were obtained through visual interpretation. Note that when interpreted, open waters and waters with small ice floes within pack ice were classified as leads, while ice and the areas in leads but with a rougher surface were classified as ice. Wind may cause water surface roughness. However, the proposed method was constrained for use in low wind speed conditions. Entropy subimages were calculated from SAR subimages using a sliding window with a size of 5 × 5 and a slide step of 1 using the following equation:

$$\tau = -\sum_{i=0}^{N} P_i \log_2 P_i \tag{1}$$

where τ represents the entropy value of the subimage, N represents the gray level in the image, and P_i represents the percentage of pixels with gray value i in the image. The more gray levels there are in the image, the richer the corresponding texture information, and the greater the image entropy [34], [35]. Since the complexity of sea water texture in SAR images

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:58:59 UTC from IEEE Xplore. Restrictions apply.

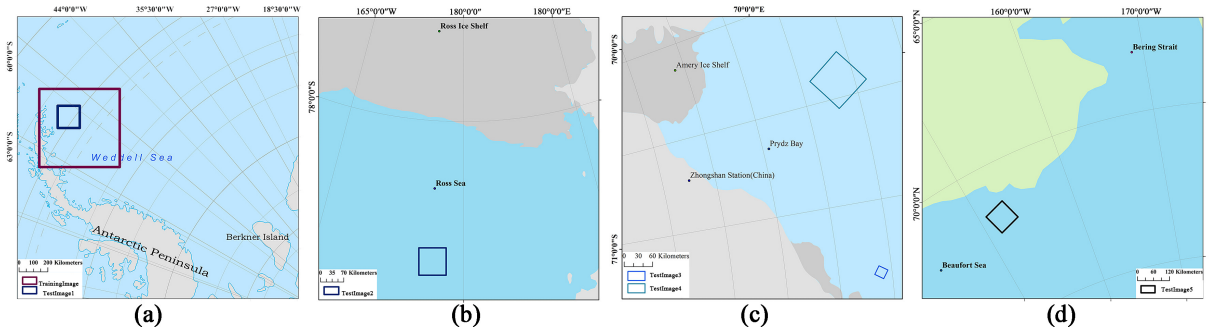


Fig. 1. Locations of the six Sentinel-1 SAR images used in this article. The deep brown box and the deep blue box in (a) denote the extent of the training image and test, respectively. The boxes in (b)–(d) show the extent of test.

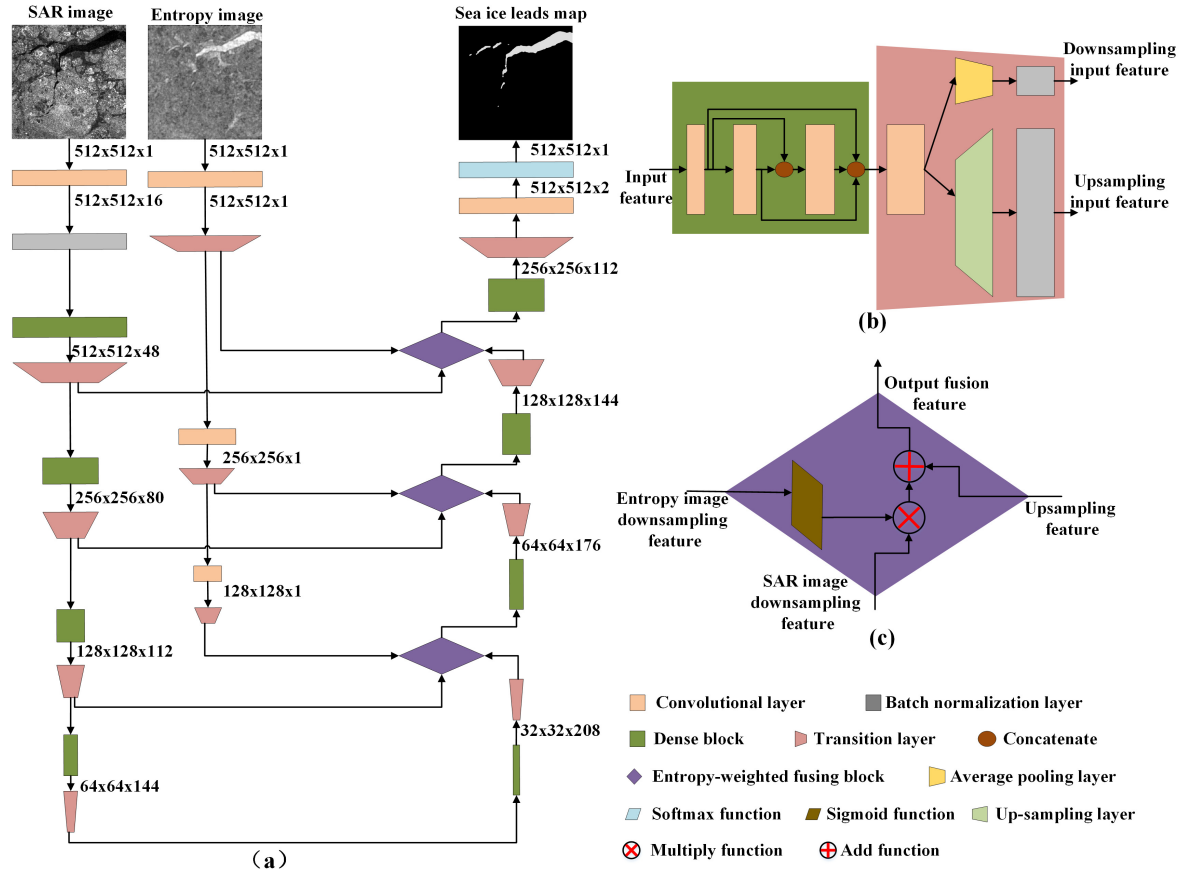


Fig. 2. Structure of EW-Net proposed in this study, where (a) describes the overall structure of the proposed EW-Net and the 3-D numbers represent the feature dimensions after the convolutional operation, (b) describes the structure of the dense block (green box) and transition layer (pink box), and (c) describes the structure of the entropy-weighted fusing block.

is less than that of sea ice, it could be assumed that there is a large difference in entropy between sea water and sea ice. Therefore, we introduced the image entropy feature into the U-Net baseline network to enhance the sensitivity of the network to the sea ice leads. A total of 3000 groups of samples were finally generated and applied in this study.

C. Entropy-Weighted Network

In this study, an entropy-weighted network (EW-Net) was proposed to detect sea ice leads from Sentinel-1 SAR images. EW-Net was mainly composed of three parts: an improved U-Net baseline network (for semantic segmentation), a pseudo-Siamese sampling structure (for generating weight), and an entropy-weighted fusing block (for image feature fusion) (see Fig. 2). The U-Net baseline network was improved by replacing the convolutional layer with a dense block, and image features of different scales were learned by performing continuous convolution and scale conversion operations. In the pseudo-Siamese sampling structure, the image was continuously downsampled as a feature map with a depth of 1, which was fed to the fusing block for feature fusion calculation. In the entropy-weighted block, three kinds of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:58:59 UTC from IEEE Xplore. Restrictions apply.

features generated from the network were weighted and added, and the fused features were returned. The input of the network was SAR images and entropy images, and the output was two-category classification maps, which were labeled "leads" and "nonleads."

The improved U-Net model in this study mainly included two parts: an image downsampling branch and a feature upsampling branch. The image downsampling branch received Sentinel-1 images with dimensions of 512 × 512 × 1, including a convolutional layer with a kernel size of 3 × 3, a batch normalization layer, and four dense blocks, which extracted remote-sensing images into highly abstract feature maps with dimensions of 32 × 32 × 144. In the feature upsampling branch, a 512 × 512 × 2 feature restoration map was generated through four dense blocks, four transition layers, and a single convolutional layer with a kernel size of 1 × 1, in which each pixel value of each dimension feature map represents the probability that the pixel was leads or nonleads. Then, the two-dimensional feature recovery map was processed by the softmax function, and the sea ice lead maps were obtained. Each dense block contained three convolutional layers with a kernel size of 3 × 3, and each layer had direct connections to all subsequent layers [see Fig. 2(b)]. The dense block can be expressed as

$$\boldsymbol{X}_l = \boldsymbol{H}_l\left(\left[\boldsymbol{X}_0, \boldsymbol{X}_1, \ldots, \boldsymbol{X}_{l-1}\right]\right) \tag{2}$$

where $\boldsymbol{l}$ represents the index of convolutional layers, $\boldsymbol{H}_l$ denotes the concatenation of all features of each layer, and $\boldsymbol{X}_l$ is the output feature map. A transition layer after the dense block contains a single convolution layer, a mean pooling layer during downsampling (or an upsampling layer during upsampling), and a batch normalization layer, which was used to adjust the dimensions of the dense block's output results to adapt to subsequent processing. As an improvement in the network, the entropy image downsampling branch was introduced into the network. The entropy image downsampling branch received the entropy images corresponding to Sentinel-1 images, including three convolution layers with 3 × 3 kernels and three transition layers with an output depth of 1 for dimensionality reduction, and extracted feature images with the same length and width synchronously with the image downsampling branch.

Another improvement in the network was the entropy-weighted fusing block [see Fig. 2(c)]. There were three inputs for the block: image features, entropy features, and upsampling features, and all of them had the same length and width. First, the range of entropy features was normalized to decimals between 0 and 1 by the sigmoid function. Next, this range was multiplied by the received image features as a weight value and then added to the received upsampling features to obtain the fused features. The structure is given in the following:

$$\boldsymbol{Y} = \text{Sigmoid}(\boldsymbol{w}) \times \boldsymbol{x} + \boldsymbol{b} \tag{3}$$

where $\boldsymbol{Y}$ is the feature map tensor after fusion, $\boldsymbol{w}$ is the entropy feature graph tensor, $\boldsymbol{x}$ is the downsampling feature map tensor, and $\boldsymbol{b}$ is the upsampling feature map tensor.

D. Network Parameter Settings and Quantitative Evaluation Indices

Before being fed into the network, SAR subimages in the sea ice lead training dataset were first normalized to the image values to speed up network fitting and then fed into the network as input 1. Entropy subimages were also normalized and fed into the network as input 2. Lead binary subimages were directly used as the label of EW-Net.

When training our network, adaptive moment estimation (Adam), one of the most commonly used algorithms, was used as the optimizer to minimize loss and update the network parameters [36]. The number of training epochs was 30, and the number of iterations in each epoch was 5000. The initial value of the learning rate was set to 0.0001 and divided by 10 every ten epochs.

To quantitatively evaluate the accuracy of the proposed method for polar sea ice lead detection, several indices based on the confusion matrix were used. The confusion matrix is usually expressed in the form of a matrix with two rows and two columns, and four indices (TP, TN, FP, and FN) represent the cell numbers of true positives, true negatives, false positives, and false negatives, respectively. According to the confusion matrix, the overall accuracy (OA), F1-score, intersection over union (IoU), and Kappa coefficient were used for quantitative evaluation [see (4)–(10)]

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{4}$$

$$\text{F1} - \text{score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{5}$$

where

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{6}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{7}$$

$$\text{IoU} = \frac{\text{TP}}{\text{FP} + \text{FN} + \text{TP}} \tag{8}$$

$$\text{Kappa} = \frac{\text{OA} - \text{Pe}}{1 - \text{Pe}} \tag{9}$$

where

$$\text{Pe} = \frac{(\text{TP} + \text{FP}) \times (\text{TP} + \text{FN}) \times (\text{FN} + \text{TN}) \times (\text{FP} + \text{TN})}{(\text{TP} + \text{FN} + \text{FP} + \text{TN})^2}. \tag{10}$$

The OA denotes the proportion of correctly classified pixels in all pixels. The F1-score is a powerful evaluation metric used to determine the harmonic mean of the precision and recall metrics. IoU calculates the ratio of the intersection and union of two classes, and Kappa is a punitive "biased" indicator used to solve the imbalance of results. The procedure for ground-truth mapping is consistent with the procedure for the production of leads in the training dataset.

III. Results

A. Comparison of Different Sea Ice Lead Detection Methods From Sentinel-1 Images of the Weddell Sea, Antarctic

Fig. 3 shows the detected sea ice leads by using our proposed EW-Net method. The boundary of sea ice leads (red

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:58:59 UTC from IEEE Xplore. Restrictions apply.

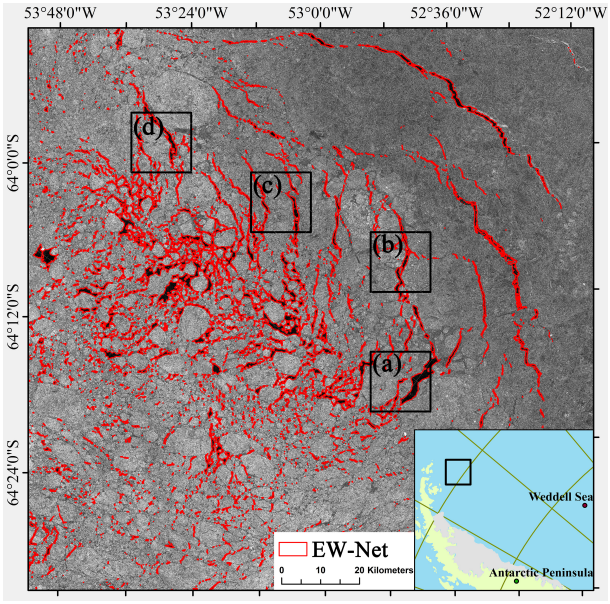


Fig. 3. Sea ice leads detected using test Sentinel-1 SAR image 1 (July 10, 2019) taking from the Weddell Sea in the Antarctic by using the EW-Net method proposed in this study. The red polygons represent the boundary of the detected sea ice leads. Four black boxes labeled (a)–(d) are the subregions selected for detailed analysis.

polygons) matched well with the dark areas of the original image, which indicates that our method detected most of the sea ice leads from the Sentinel-1 SAR image.

Four subregions in Fig. 3 were selected to quantitatively evaluate the accuracy of the EW-Net method. Sea ice lead detection based on the traditional threshold classification method and two commonly used deep learning methods, U-Net [25] and DeepLab V3 Plus [37], were compared with EW-Net. It should be noted that the U-Net and DeeplLab V3 Plus methods applied the same sea ice lead training dataset. Fig. 4 shows the comparison of the detected sea ice leads from the four methods mentioned above.

Sea ice leads obtained by the threshold method accurately detected the edge of the leads and the detailed information of sea ice leads; however, a large number of patches unrelated to leads were also detected, which may be caused by the melting ice or the noise of the SAR image. Some shallow black regions, such as the area of the black circle in the threshold result in Fig. 4(d), were detected by mistake. Sea ice leads from the DeepLab V3 plus method performed the worst among the four methods because of the loss of detailed information on the bodies of main leads. The result from the U-Net method was considerably improved; however, holes still existed in the detected sea ice leads due to the influence of some small ice floes, as shown in the area of the black circle in the U-Net results in Fig. 4(c) and (d). Sea ice leads obtained from the proposed EW-Net methods, which exhibited high stability, were the closest to the ground truth and maintained good performance on lead detection regardless of the integrity of the leads or the influence of SAR image noise.

A quantitative evaluation of the four different sea ice lead detection methods (see Table II) shows that the EW-Net method has advantages in all five evaluation indices, which indicates that the proposed method in this article exhibits better sea ice lead detection performance than the other three methods.

TABLE II

ACCURACY OF SEA ICE LEAD DETECTED FROM DIFFERENT DETECTION METHODS OF THRESHOLD, U-NET, DEEPLAB V3 PLUS, AND EW-NET

Method	Region	OA	F1-score	IoU	Kappa
Threshold	(a)	0.9779	0.9115	0.8374	0.8989
	(b)	0.9833	0.8317	0.7120	0.8230
	(c)	0.9776	0.8635	0.7597	0.8513
	(d)	0.9698	0.8256	0.7030	0.8091
Mean		0.9771	0.8581	0.7530	0.8456
U-Net	(a)	0.9903	0.9611	0.9252	0.9556
	(b)	0.9855	0.8460	0.7331	0.8384
	(c)	0.9861	0.9129	0.8397	0.9053
	(d)	0.9804	0.8716	0.7724	0.8610
Mean		0.9856	0.8979	0.8176	0.8901
DeepLab V3 Plus	(a)	0.8872	0.1455	0.0785	0.1299
	(b)	0.9786	0.7167	0.5585	0.7061
	(c)	0.9426	0.4751	0.3115	0.4532
	(d)	0.9445	0.4976	0.3312	0.4757
Mean		0.9382	0.4587	0.3199	0.4412
EW-Net	(a)	0.9980	0.9919	0.9840	0.9908
	(b)	0.9825	0.8301	0.7095	0.8209
	(c)	0.9850	0.9093	0.8337	0.9012
	(d)	0.9941	0.9632	0.9290	0.9600
Mean		**0.9899**	**0.9236**	**0.8641**	**0.9182**

B. Comparison of Different Sea Ice Lead Detection Methods From Sentinel-1 Images of the Ross Sea, Antarctica

We also performed tests and experiments on the generalizability of the proposed method to other images of areas other than the Weddell Sea. A Sentinel-1 image of the Ross Sea was used for the applicability test. This image had a size of 2480 × 2480 and was never used for neural network training. The detected lead maps of U-Net, DeepLab V3 plus, and the threshold are compared with the results of our method again. The test image with the lead map detected by EW-Net is shown in Fig. 5.

It is obvious that most of the leads are detected from the image taken of the Ross Sea. It seems that the thinner leads, whose width is less than 3 pixels on the image, are not detected, but this may not be attributed to the EW-Net model proposed in this article. There are two reasons for this phenomenon: 1) when the sea ice leads dataset for training is generated by visual interpretation, it is difficult to draw the boundary of all leads perfectly and some leads pixels will inevitably be lost and 2) the principles we follow when generating lead datasets allow very fine leads to not be classified as leads. However, all three deep learning networks participating in the comparison are trained based on the same

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:58:59 UTC from IEEE Xplore. Restrictions apply.

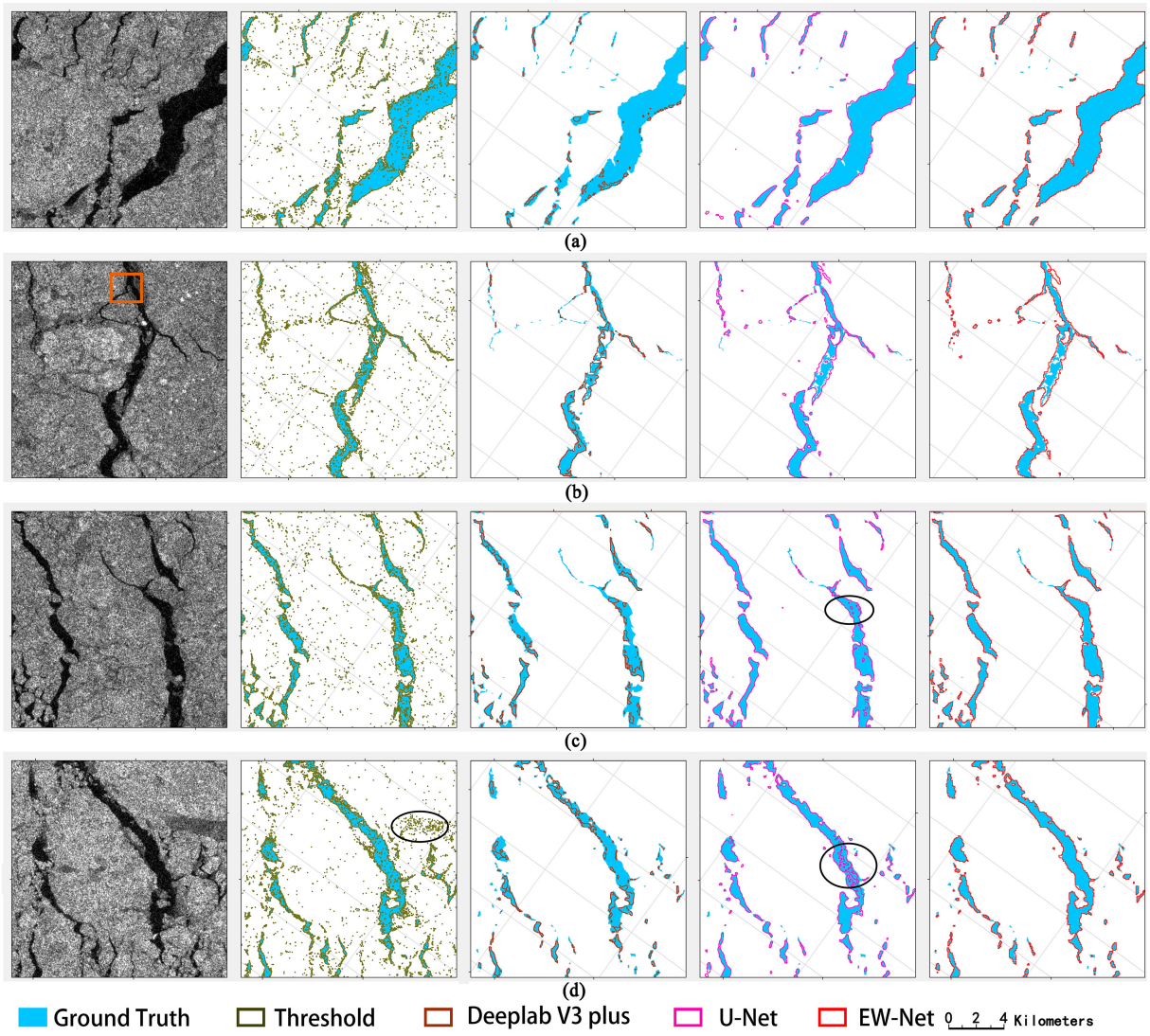


Fig. 4. Comparison of sea ice leads detected from the threshold, U-Net, DeepLab V3 Plus, and EW-Net methods, where (a)–(d) represent the detected sea ice leads from four subregion images selected from Fig. 3 of the Weddell Sea in the Antarctic. The blue area denotes the ground truth, and the polygons with different colors represent the sea ice leads detected from four comparative methods.

sea ice leads dataset, and the test results produced by such experiments are considered to be comparable. Similarly, the subimages of the two regions in the image are selected for further comparison and precision calculation. The comparison results are shown in Fig. 6.

As shown in Fig. 6, there are obviously two parts in the SAR image: a bright area and dark area, and there are few shallow dark areas that do not represent leads that affect the results. In this case, both U-Net and EW-Net have good and similar lead map results. When observing the details, it was found that the results of EW-Net are more consistent with the ground truth, as shown in the black circle of the EW-Net result in Fig. 6(e). The lead map of the threshold method always had the most details. Nevertheless, the threshold method classified more areas covered with a large amount of thin ice as leads, as shown in the black circle of the threshold result in Fig. 6(e), which is an undesirable result. The leads map of DeepLab V3 plus lost a large amount of lead information. The quantitative evaluation of these two subgraphs and the accuracy results are shown in Table III. EW-Net has the optimal average value of each index, and the threshold method detects the most potential lead information, although there are many commission errors. The indices of DeepLab V3 plus are still the worst. U-Net and EW-Net have very close values in each index, which indicates that U-Net and EW-Net have similar performance for SAR images with clear distinctions between ice and water.

C. Comparison of Different Sea Ice Lead Detection Methods From Sentinel-1 Images of Prydz Bay, Antarctica

In this section, two Sentinel-1 images near the vicinity of Prydz Bay, Antarctica, are tested. Both images have a size of 2048 × 2048 and have never been used for neural network training. The difference between the two images is that the acquisition mode of test image 3 was acquired using the interferometric wide (IW) swath mode, while test image 4 was acquired using the extra wide (EW) swath mode. The EW

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:58:59 UTC from IEEE Xplore. Restrictions apply.

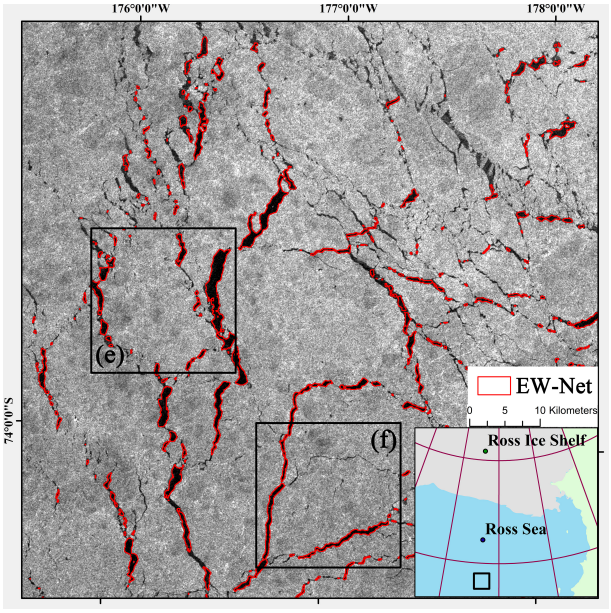


Fig. 5. Sea ice leads detected from test Sentinel-1 SAR image 2 (May 30, 2020) of the Ross Sea, Antarctica, by using the EW-Net method proposed in this study. The red polygons represent the boundary of the detected sea ice leads. The two black boxes labeled (e) and (f) are subregions selected for detailed analysis.

TABLE III

ACCURACY OF THE SEA ICE LEADS DETECTION METHODS OF THE THRESHOLD, U-NET, DEEPLAB V3 PLUS, AND EW-NET

Method	Num	OA	F1-score	IoU	Kappa
Threshold	(e)	0.9733	0.8671	0.7654	0.8525
	(f)	0.9856	0.8438	0.7298	0.8364
Mean		0.9794	0.8555	0.7476	0.8445
Deeplab V3 Plus	(e)	0.9144	0.0545	0.0280	0.0499
	(f)	0.9723	0.4561	0.2954	0.4461
Mean		0.9434	0.2553	0.1617	0.2480
U-Net	(e)	0.9766	0.8638	0.7603	0.8510
	(f)	0.9906	0.8914	0.8041	0.8866
Mean		0.9836	0.8776	0.7822	0.8688
EW-Net	(e)	0.9823	0.8962	0.8120	0.8865
	(f)	0.9907	0.8885	0.7994	0.8837
Mean		**0.9865**	**0.8924**	**0.8057**	**0.8851**

mode acquires data over a 400-km swath with a 20 m × 40 m spatial resolution, while the IW mode acquires data with a 250-km swath with a 5 m × 20 m spatial resolution. The sea ice lead maps obtained from the two view images via EW-Net are shown in Figs. 7 and 8.

From both images, it is easier to visually identify sea ice leads in the images acquired from the IW mode than from the EW mode due to the improved spatial resolution. The results in Fig. 7 show that the detected lead boundaries are relatively complete, with a few omissions, but there are more complex boundaries in the wider leads, as in the bottom-right corner,

TABLE IV

ACCURACY OF SEA ICE LEAD DETECTED FROM DIFFERENT DETECTION METHODS OF THRESHOLD, U-NET, DEEPLAB V3 PLUS, AND EW-NET

Method	Region	OA	F1-score	IoU	Kappa
Threshold	(g)	0.9802	0.8987	0.8160	0.8878
	(h)	0.9670	0.8958	0.8113	0.8762
	(i)	0.9859	0.8781	0.7827	0.8707
	(j)	0.9609	0.8801	0.7858	0.8567
Mean		0.9735	0.8882	0.7989	0.8728
DeepLab V3 Plus	(g)	0.9228	0.2519	0.1441	0.2343
	(h)	0.8697	0.2942	0.1724	0.2591
	(i)	0.9850	0.8743	0.7767	0.8663
	(j)	0.9280	0.7364	0.5828	0.6966
Mean		0.9264	0.5392	0.4190	0.5141
U-Net	(g)	0.9704	0.8154	0.6883	0.7995
	(h)	0.9280	0.7280	0.5723	0.6881
	(i)	0.9839	0.8615	0.7567	0.8530
	(j)	0.9402	0.7892	0.6518	0.7553
Mean		0.9556	0.7985	0.6673	0.7740
EW-Net	(g)	0.9841	0.9069	0.8297	0.8983
	(h)	0.9758	0.9208	0.8533	0.9065
	(i)	0.9850	0.8816	0.7882	0.8736
	(j)	0.9619	0.8820	0.7888	0.8593
Mean		**0.9767**	**0.8978**	**0.8150**	**0.8844**

which reveals a direction for improvement in this method. The leads in the map in Fig. 8 are relatively good except for a few omissions in the top-left corner. Details of the lead detection results for the four subregions using the four methods are compared in Fig. 9.

As shown in Fig. 9, the method in this article is able to extract the boundaries of sea ice leads relatively completely and with minimal misdetection, regardless of whether they are linearly curved or short and fragmented. The omissions in the middle of the leads in Fig. 9(g) and (h) suggest that there is room for improvement. The threshold method has the highest false detection rate, as this rule-based classification method is more subject to noise interference in the SAR classification task. The two deep learning methods, DeepLab V3 plus and U-Net, are largely immune to noise, as is EW-Net, but result in more missed data. These results are further supported by the quantitative evaluation data in Table IV. Overall, EW-Net is also generalizable for images acquired in the IW mode.

D. Comparison of Different Sea Ice Lead Detection Methods From Sentinel-1 Image in the Beaufort Sea, Arctic

We also performed applicability tests of the proposed method on another image located in the Beaufort Sea in the Arctic. The new SAR image from Sentinel-1 has a size of 2480 × 2480 and has never been used for neural network training. Fig. 10 shows the detected sea ice leads from EW-Net, and Fig. 11 shows the comparison of sea ice leads

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:58:59 UTC from IEEE Xplore. Restrictions apply.

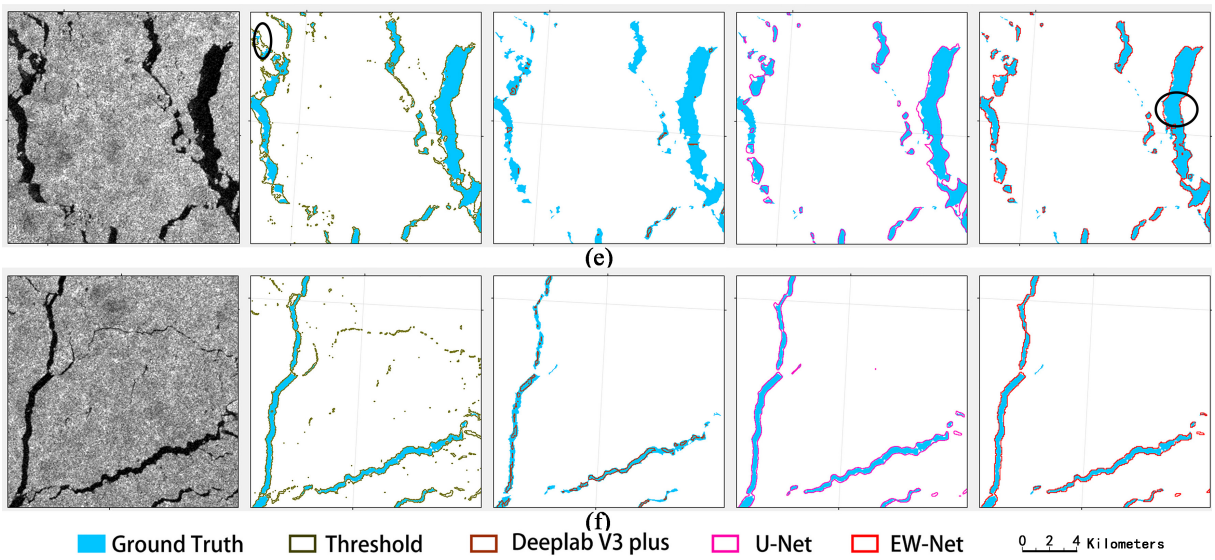


Fig. 6. Comparison of sea ice leads detected from the threshold, U-Net, DeepLab V3 Plus, and EW-Net methods, where (e) and (f) represent the detected sea ice leads from two subregion images of the Ross Sea, Antarctica, selected from Fig. 5. The blue area denotes the ground truth, and the polygons with different colors represent the sea ice leads detected from the four comparative methods.

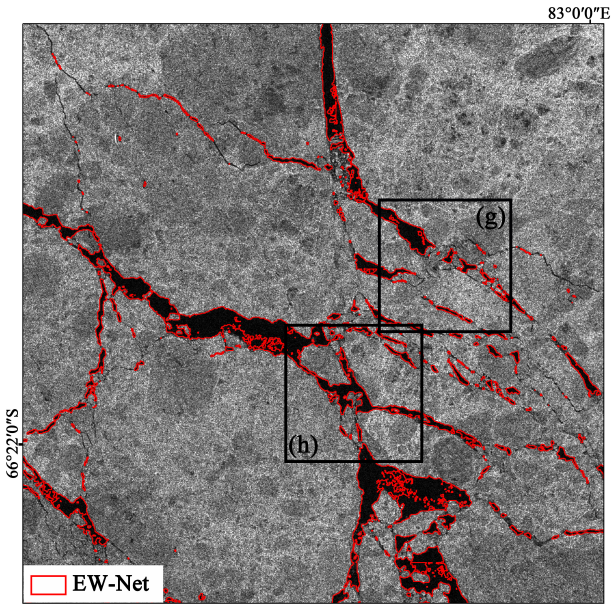


Fig. 7. Sea ice leads detected by the IW swath mode from test Sentinel-1 SAR image 3 (February 6, 2021) in Prydz Bay, Antarctica, by using the EW-Net method proposed in this study. The red polygons represent the boundary of the detected sea ice leads. The two black boxes labeled (g) and (h) are the subregions selected for detailed analysis.

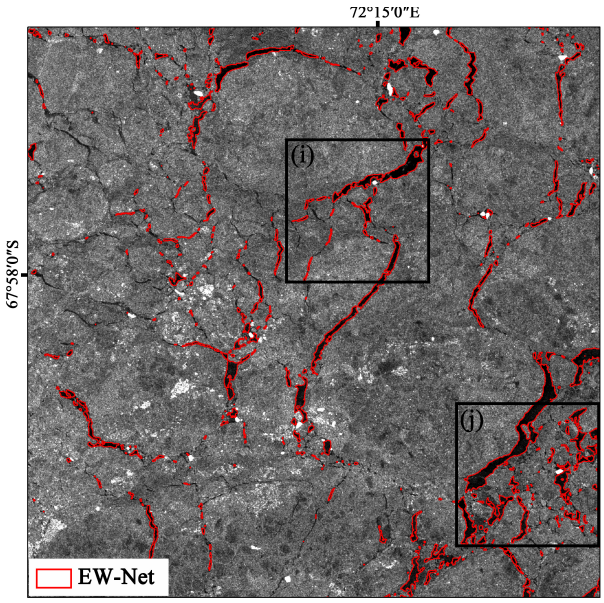


Fig. 8. Sea ice leads detected from test Sentinel-1 SAR image 4 (November 3, 2020) in Prydz Bay, Antarctica, by using the EW-Net method proposed in this study. The red polygons represent the boundary of the detected sea ice leads. Two black boxes labeled (i) and (j) are the subregions selected for detailed analysis.

from the threshold, U-Net, DeepLab V3 Plus, and EW-Net methods using the subregion images of Fig. 10(k) and (l).

The dark areas in Fig. 10 can be divided into deep dark areas and shallow dark areas, which represent open waters and frozen waters, respectively. The red polygons covered most of the deep dark areas, which indicates that the EW-Net method could detect the true sea ice leads.

From the SAR images in Fig. 11, the deep dark area and the shallow dark area in the leads can be clearly observed, and when extracting the ground-truth image of sea ice leads, only the deep dark areas should be covered as much as possible. More areas covered by ice were detected by the threshold method, which presented low precision (see Table V). It was difficult to detect sea ice leads by using the DeepLab V3 Plus method, and all the values of the evaluation index in Table V were the lowest by using this method. Sea ice leads detected by using the U-Net method lost a large amount of detailed information, while sea ice leads detected by the EW-Net method proposed in this article matched well with the ground truth and exhibited the best performance among the comparative methods for sea ice lead detection.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:58:59 UTC from IEEE Xplore. Restrictions apply.

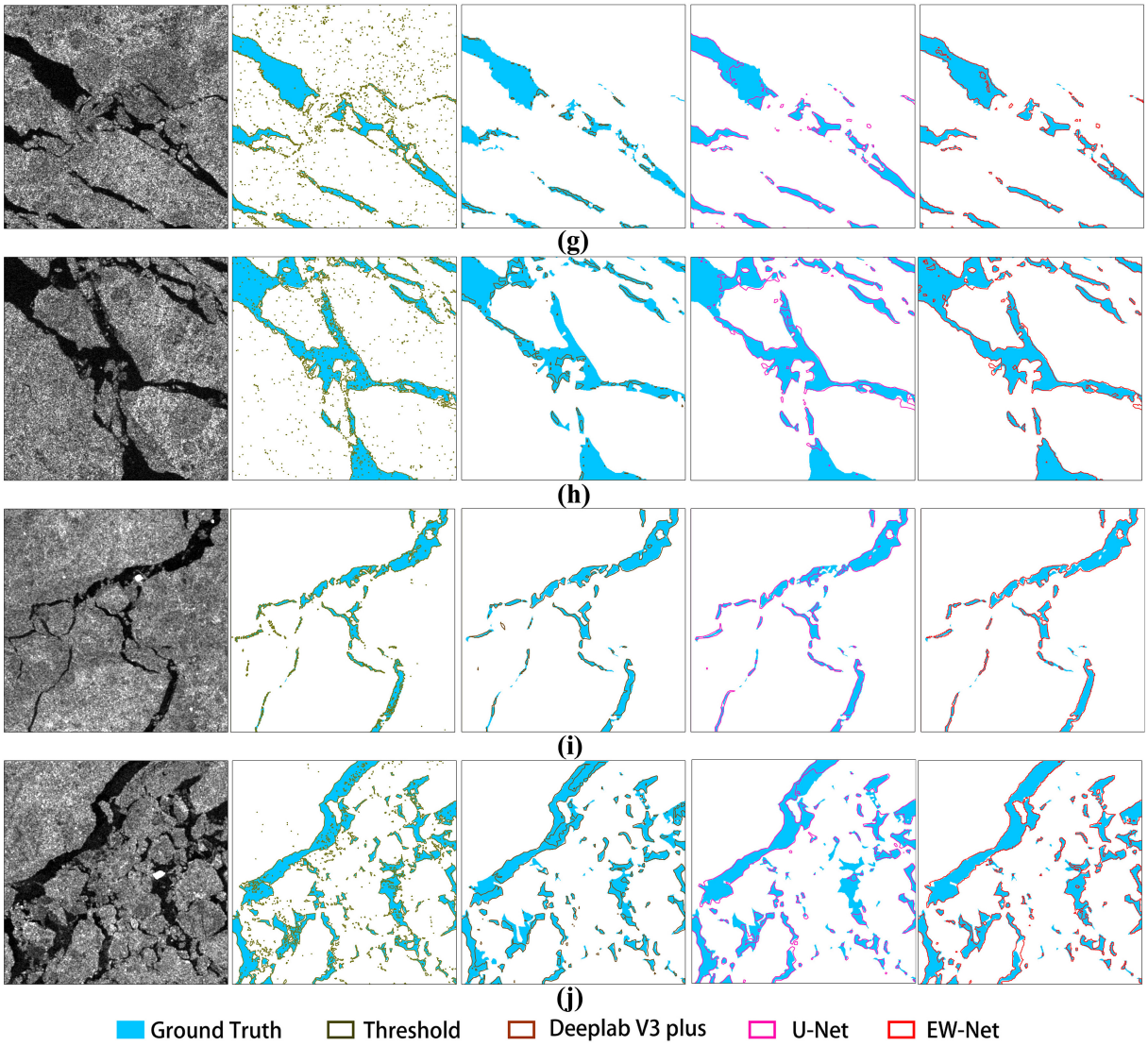


Fig. 9. Comparison of sea ice leads detected from the threshold, U-Net, DeepLab V3 Plus, and EW-Net methods, where (g)–(j) represent the detected sea ice leads from four subregion images in Prydz Bay, Antarctica, selected from Figs. 7 and 8. The blue area denotes the ground truth, and the polygons with different colors represent the sea ice leads detected by the four comparative methods.

IV. Discussion

In this study, a novel EW-Net based on deep learning was proposed to automatically detect polar sea ice leads from Sentinel-1 SAR images, which was explored to address traditional image classification methods that are affected by SAR image noise and poor generalization. All the sea ice lead detection experiments in the Antarctic and the Arctic regions in this article proved the high accuracy and effectiveness of the proposed EW-Net method for polar sea ice lead detection.

In the SAR imaging process, the gray values of adjacent pixels in the radar echo signal will have random variations due to coherence, and these variations will lead to speckle noise in the image [21]. Thus, it is challenging to discriminate leads and ice accurately from SAR images. From the results of our experiments, rule-based classification methods, such as the threshold method, were not applicable because of the similar value of backscattering between the noise and leads. The sea ice lead detection method based on deep learning was minimally affected by SAR image noise because of its ability to learn features according to the training data. Therefore, we recommend detecting sea ice leads by using the deep learning method.

One of the goals of our work for polar sea ice lead detection was to avoid complex SAR image preprocessing and directly to use the original images to obtain accurate polar sea ice lead information. It should be noted that none of the SAR images in the experiments in this study underwent preprocessing, such as radiometric calibration. The detection of leads based on the threshold method requires setting the appropriate value for each image and dividing the image into two categories. This method was affected by subjective factors and could not detect leads on SAR images automatically. Fig. 4 and Table II show that the EW-Net method was completely feasible for sea ice detection of SAR images without preprocessing, which

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:58:59 UTC from IEEE Xplore. Restrictions apply.

TABLE V

ACCURACY OF SEA ICE LEAD DETECTED FROM DIFFERENT DETECTION METHODS OF THRESHOLD, U-NET, DEEPLAB V3 PLUS, AND EW-NET

Method	Region	OA	F1-score	IoU	Kappa
Threshold	(k)	0.9159	0.6153	0.4443	0.5751
	(l)	0.9762	0.7750	0.6326	0.7626
Mean		0.9461	0.6952	0.5385	0.6689
DeepLab V3 Plus	(k)	0.9375	0.1795	0.0986	0.1690
	(l)	0.9537	0.0012	0.0006	0.0011
Mean		0.9456	0.0904	0.0496	0.0851
U-Net	(k)	0.9452	0.4353	0.2782	0.4119
	(l)	0.9707	0.7470	0.5962	0.7321
Mean		0.9580	0.5912	0.4372	0.5720
EW-Net	(k)	0.9686	0.7769	0.6352	0.7600
	(l)	0.9773	0.7932	0.6572	0.7815
Mean		**0.9730**	**0.7851**	**0.6462**	**0.7708**

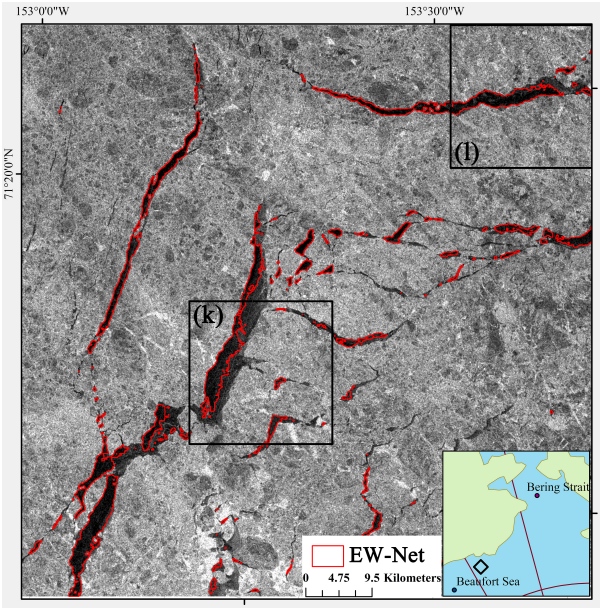


Fig. 10. Sea ice leads detected from test Sentinel-1 SAR image 5 (March 31, 2020) of the Beaufort Sea in the Arctic by using the proposed EW-Net method in this study. The red polygons are the sea ice lead boundaries detected by EW-Net. Two black box areas labeled (k) and (l) were the subregions selected for detailed analysis.

TABLE VI

ERRORS OF SEA ICE LEAD DETECTED FROM DIFFERENT METHODS OF THE THRESHOLD, U-NET, DEEPLAB V3 PLUS, AND EW-NET

Method	Commission error		Omission error		Patch number of leads*
	pixel	percent	pixel	percent	
Threshold	55	7.28	127	15.36	1+38
DeepLab V3 Plus	453	60.00	10	3.21	1+5
U-Net	332	43.97	11	2.53	1+4
EW-Net	14	1.85	184	19.89	1+**1**

* The number on the left of "+" indicates the patch number of ground truth, and the sum indicates the patch number of detected leads. The number on the right represents the difference between detected leads and ground truth, and the smaller the better.

TABLE VII

ACCURACY OF SEA ICE LEADS DETECTION METHODS OF THE THRESHOLD, U-NET, DEEPLAB V3 PLUS, AND EW-NET

Method	Region	OA	F1-score	IoU	Kappa	Patch number of leads*
Threshold	(m)	0.9080	0.6783	0.5132	0.6268	18+570
	(n)	0.9073	0.7122	0.5530	0.6592	4+552
Mean		0.9076	0.6952	0.5331	0.6430	
DeepLab V3 Plus	(m)	0.8996	0.6026	0.4312	0.5527	18+74
	(n)	0.8727	0.5148	0.3466	0.4588	4+71
Mean		0.8861	0.5587	0.3889	0.5058	
U-Net	(m)	0.9260	0.7509	0.6012	0.7085	18+65
	(n)	0.8565	0.4279	0.2722	0.3710	4+76
Mean		0.8913	0.5894	0.4367	0.5397	
EW-Net	(m)	0.9705	0.9167	0.8462	0.8988	18+**25**
	(n)	0.9836	0.9574	0.9182	0.9472	4+**16**
Mean		**0.9770**	**0.9370**	**0.8822**	**0.9230**	

* The number on the left of "+" indicates the patch number of ground truth, and the sum indicates the patch number of detected leads. The number on the right represents the difference between detected leads and ground truth, and the smaller the better.

benefits from two advantages of deep learning: high degrees of freedom for input data and the self-learning ability of image features [22]. Therefore, this approach can be a promising and effective way to improve the efficiency of rapid mapping of polar sea ice leads by minimizing image processing steps while ensuring high accuracy.

Generally, omission error and commission error are the two indices of concern in image classification. Fig. 12 shows an enlarged view of the orange box area in Fig. 4(b) and shows the comparison of the omission error and the commission error in the lead detection results of these different methods. Table VI shows the statistical results of the errors of detecting leads through threshold, U-Net, DeepLab V3 Plus, and EW-Net methods. The pixel size of the selected area is 61 × 57, in which the number of lead patches is 1 and the number of its pixels is 755. The selected region contains obvious sea ice leads, which have two branches: the left branch is narrower and the right branch is wider. All areas other than the leads are ice. The results of the threshold method contain more pink pixels (commission error) in the ice area and fewer yellow pixels (omission error) in lead areas, which shows that this method easily and incorrectly classified dark pixels on ice as leads, while relatively bright pixels in leads were easily divided into nonleads, depending on the selection of the threshold. The results of DeepLab V3 Plus and U-Net

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:58:59 UTC from IEEE Xplore. Restrictions apply.

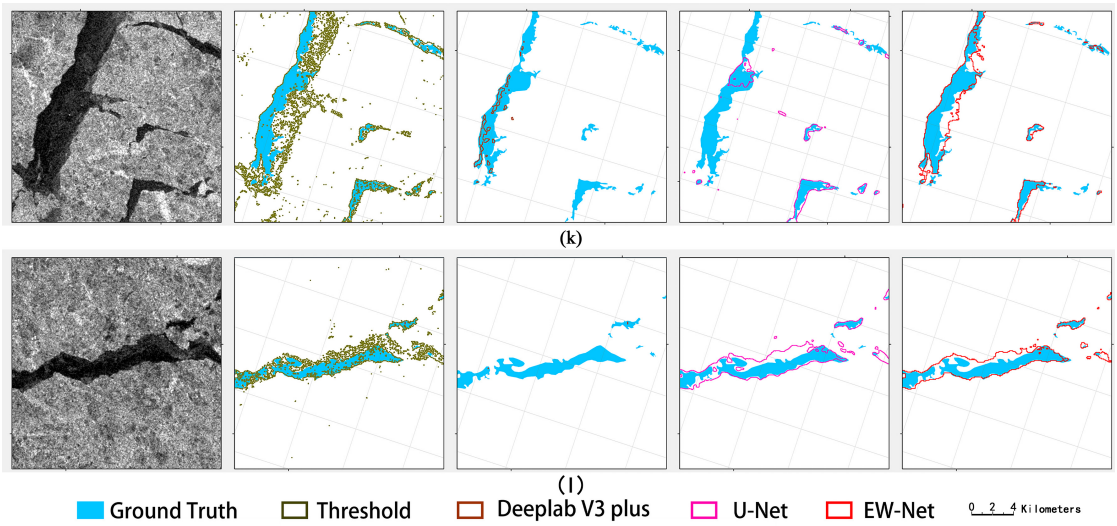


Fig. 11. Comparison of sea ice leads detected from the threshold, U-Net, DeepLab V3 Plus, and EW-Net methods, where (k) and (l) represent the detected sea ice leads from two subregion images of the Beaufort Sea in the Arctic selected from Fig. 10. The blue area denotes the ground truth, and the polygons with different colors represent the sea ice leads detected by the four comparative methods.

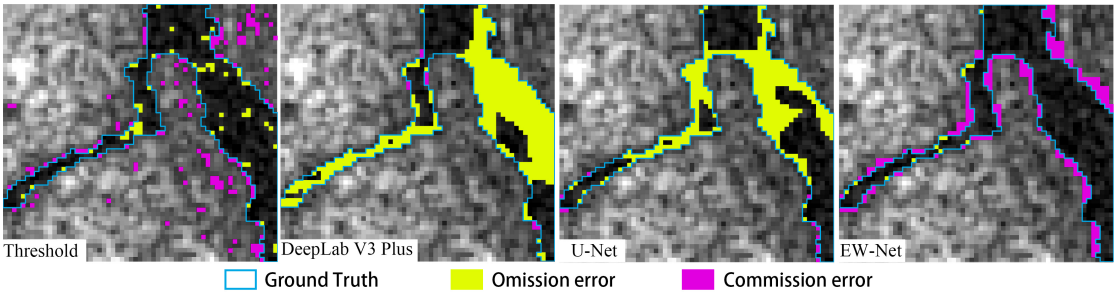


Fig. 12. Comparison of error detection of sea ice leads by the threshold, U-Net, DeepLab V3 Plus, and EW-Net methods. The image is an enlarged view of the orange box region in Fig. 4(b) and has a size of 61 × 57. The blue polygons represent the ground truth of sea ice leads, the yellow pixels represent the commission error, and the pink pixels represent the omission error.

show that there are many omission errors in the lead areas, which is not the desired result. The network structure of DeepLab V3 Plus is complex, but it cannot fit well when the number of samples is small. The feature sampling depth of U-Net is insufficient, and the feature fusion method is simple, which leads to the easy loss of image information. There are very few yellow pixels in the result of EW-Net, which demonstrates that our method rarely misses leads and has high integrity. However, there are pink pixels at the junction of leads and ice, which shows that this method has room for improvement in lead edge detection. EW-Net takes U-Net as the baseline network, increases sampling depth by introducing the dense block [37], enhances lead information by adding an entropy branch, and improves feature utilization by using an entropy-weighted fusion block [34], [38], achieving the desired results. The data in the columns "commission error" and "omission error" in Table IV provide the errors of each method from a quantitative point of view. Compared with other methods, the omission error of our method only accounts for 1.85% of the number of true lead pixels. Although the number of commission errors accounts for almost 20% of the total number of detected lead pixels, EW-Net is relatively more applicable than the other methods. It is worth noting that the data in the "number of lead patches" column show a comparison of patch numbers between detected leads and leads in the ground truth. To compare with the ground truth more clearly, an addition expression is used to express the difference with the true value, in which the number before the plus sign indicates the patch number of the ground truth,

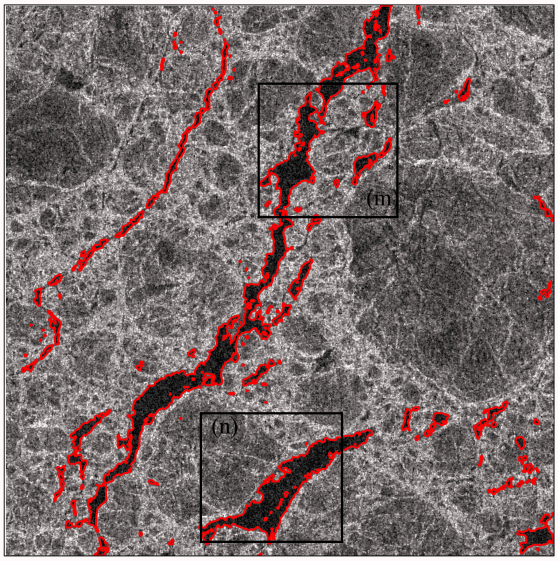


Fig. 13. Sea ice leads detected from the ALOS PALSAR image (March 15, 2011) of the Weddell Sea in the Antarctic by using the proposed EW-Net method. The red polygons are the lead boundaries detected by EW-Net. Two black boxes labeled (m) and (n) were the subregions selected for detailed analysis.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:58:59 UTC from IEEE Xplore. Restrictions apply.

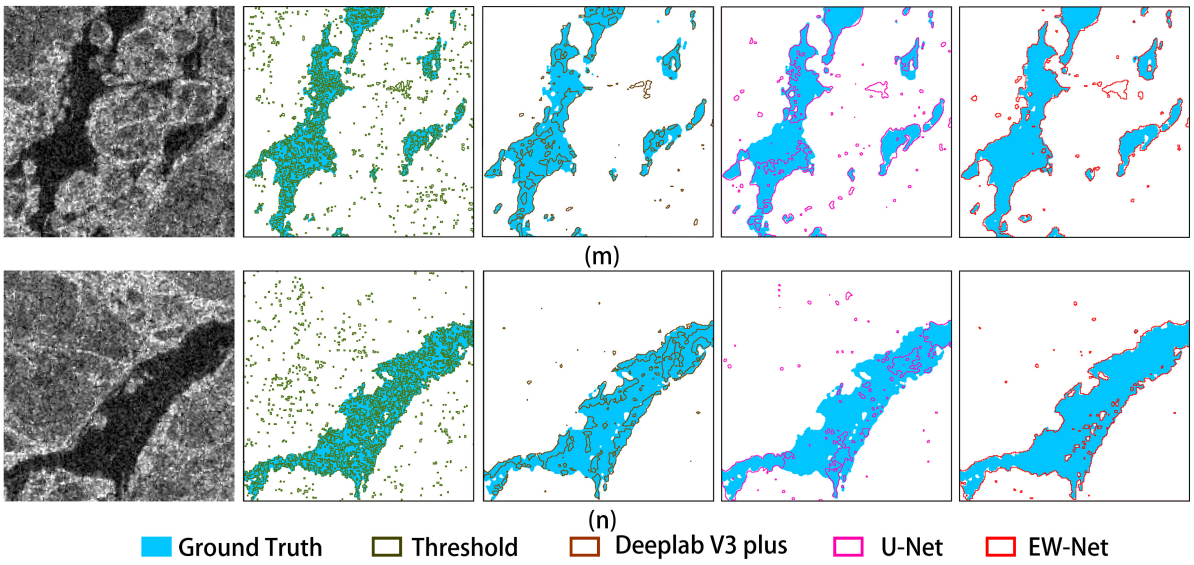


Fig. 14. Comparison of sea ice leads detected from the untrained ALOS PALSAR image by the threshold, U-Net, DeepLab V3 plus, and EW-Net methods, where (m) and (n) represent the sea ice leads from two subregion images of the Weddell Sea in the Antarctic selected from Fig. 13. The blue area denotes the ground truth, and the polygons with different colors represent the sea ice leads detected from the four comparative methods.

and the sum of additions is the patch number of the detected leads. The threshold has a large patch number due to the high fragmentation, and the data of our method are very close to the ground truth.

Generalization is another focus of image classification algorithms, especially in the case of cross sensors. We applied the proposed method to the detection of sea ice leads from ALOS PALSAR data and compared the results with those of other methods. ALOS PALSAR is an SAR aboard the Earth observation satellite ALOS launched by Japan on January 24, 2006. The ALOS PALSAR data are quite different from the Sentinel-1 data. Sentinel-1 carries a single C-band SAR instrument operating at a center frequency of 5.405 GHz, while ALOS PALSAR works in the L-band with a center frequency of 1.27 GHz. It has been shown that L-band SAR has a penetration depth up to tens of centimeters, while the penetration depths into new sea ice for C-band SAR are on the order of centimeters, and the intensity range covered by the different ice types was larger at the L-band than at the C-band [38], [39]. The ALOS PALSAR image used in this experiment is located in the Weddell Sea of the Antarctic (March 15, 2011). The image has a size of 2048 $\times$ 2048 and was not preprocessed. Fig. 13 shows the result of leads detected through EW-Net. Overall, EW-Net could detect almost all observable leads.

Fig. 14 shows the comparison of the lead detection of the two selected areas. Although the threshold value is not universal, we still present the detection results for comparison. Similar to the previous results, the leads detected by the threshold method show the characteristics of broken patches, which indicates that the threshold method is difficult to further apply. DeepLab V3 Plus detects more leads than before, but there are still many omissions and a few commissions. Different from the previous experimental results, the lead maps detected by U-Net are much worse than those detected by EW-Net. It can be seen from the results of U-Net that there are more defects in leads. In contrast, EW-Net could detect sea ice leads better, and the edge of leads matches the ground truth well. The quantitative statistics (see Table VII) in the generalization performance verification experiment confirm that the proposed method in this study is at least 7% points higher than other methods in the OA index, at least 20% higher in F1-score, at least 30% higher in IoU, and at least 25% higher in Kappa, which demonstrates that our method has better generalization than other comparative methods.

V. Conclusion

In this study, aiming at addressing the problem of noise and poor generalization of rule-based classification methods, an EW-Net was proposed to detect polar sea ice leads from Sentinel-1 SAR images under low wind speed conditions. In EW-Net, a dense block was introduced into the U-Net baseline network to strengthen feature propagation, and an entropy sampling structure, whose features were used as a weight, was designed to improve the pertinence of the network to the leads and alleviate the influence of SAR image noise. Furthermore, an entropy-weighted feature fusion block was designed for better feature recovery. Experiments on sea ice lead detection from Sentinel-1 SAR images showed that the sea ice lead map detected through EW-Net is consistent with the ground truth in different regions at different times and spatial resolutions and is rarely affected by noise. In particular, in the cross-sensor image generalization experiment, the proposed EW-Net detected sea ice leads more effectively from ALOS PALSAR images compared to the four methods. The OA of EW-Net in all the experiments exceeds 97%, which indicates that our method has a more valuable generalization ability.

Compared with the traditional threshold method and the two classic deep learning methods U-Net and DeepLab V3 Plus, the proposed EW-Net method can directly use original SAR

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:58:59 UTC from IEEE Xplore. Restrictions apply.

images without complex preprocessing and is less affected by noise. EW-Net has shown advantages in detecting integral linear lead characteristics and capturing more scale lead information and has better applicability to cross-sensor images. However, the low wind speed constraint limits the applicability of EW-Net, which will be a breakthrough point for this method in future research.

The proposed EW-Net method is a promising and effective method to automatically obtain highly accurate polar sea ice lead information, which could benefit related research on polar sea ice, marine ecosystems, and the safety of polar ship navigation.

REFERENCES

[1] J. Hoffman, S. Ackerman, Y. Liu, and J. Key, "The detection and characterization of Arctic sea ice leads with satellite imagers," *Remote Sens.*, vol. 11, no. 5, p. 521, Mar. 2019, doi: 10.3390/rs11050521.

[2] S. Marcq and J. Weiss, "Influence of sea ice lead-width distribution on turbulent heat transfer between the ocean and the atmosphere," *Cryosphere*, vol. 6, no. 1, pp. 143–156, Feb. 2012, doi: 10.5194/tc-6-143-2012.

[3] R. Kwok, "Deformation of the Arctic Ocean sea ice cover between November 1996 and April 1997: A qualitative survey," in *Proc. Symp. Scaling Laws Ice Mech. Ice Dyn. (IUTAM)*, Apr. 2001, pp. 315–322, doi: 10.1007/978-94-015-9735-7_26.

[4] M. Qu, X. Pang, X. Zhao, J. Zhang, Q. Ji, and P. Fan, "Estimation of turbulent heat flux over leads using satellite thermal images," *Cryosphere*, vol. 13, no. 6, pp. 1565–1582, 2019, doi: 10.5194/tc-13-1565-2019.

[5] G. A. Maykut, "Energy exchange over young sea ice in the central Arctic," *J. Geophys. Res.*, vol. 83, no. C7, pp. 3646–3658, 1978, doi: 10.1029/jc083ic07p03646.

[6] E. Ólason, P. Rampal, and V. Dansereau, "On the statistical properties of sea ice lead fraction and heat fluxes in the Arctic," *Cryosph. Discuss.*, vol. 20, pp. 1–19, Jan. 2020, doi: 10.5194/tc-2020-13.

[7] P. Assmy *et al.*, "Leads in Arctic pack ice enable early phytoplankton Blooms below snow-covered sea ice," *Sci. Rep.*, vol. 7, no. 1, pp. 1–9, Dec. 2016, doi: 10.1038/srep40850.

[8] C. W. Moore *et al.*, "Convective forcing of mercury and ozone in the Arctic boundary layer induced by leads in sea ice," *Nature*, vol. 506, no. 7486, pp. 81–84, Feb. 2014, doi: 10.1038/nature12924.

[9] N. Longepe *et al.*, "Comparative evaluation of sea ice lead detection based on SAR imagery and altimeter data," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 6, pp. 4050–4061, Jun. 2019, doi: 10.1109/TGRS.2018.2889519.

[10] T. Xie, W. Perrie, C. Wei, and L. Zhao, "Discrimination of open water from sea ice in the Labrador Sea using quad-polarized synthetic aperture radar," *Remote Sens. Environ.*, vol. 247, Apr. 2020, Art no. 111948, doi: 10.1016/j.rse.2020.111948.

[11] M. Imani and H. Ghassemian, "An overview on spectral and spatial information fusion for hyperspectral image classification: Current trends and challenges," *Inf. Fusion*, vol. 59, pp. 59–83, Oct. 2020, doi: 10.1016/j.inffus.2020.01.007.

[12] Y. Dong, T. Liang, Y. Zhang, and B. Du, "Spectral–spatial weighted kernel manifold embedded distribution alignment for remote sensing image classification," *IEEE Trans. Cybern.*, vol. 51, no. 6, pp. 3185–3197, Jun. 2021, doi: 10.1109/TCYB.2020.3004263.

[13] F. Gao, X. Wang, Y. Gao, J. Dong, and S. Wang, "Sea ice change detection in SAR images based on convolutional-wavelet neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 8, pp. 1240–1244, Aug. 2019, doi: 10.1109/LGRS.2019.2895656.

[14] R. M. Haralick, K. Shanmugam, and I. Dinstein, "Textural features for image classification," *IEEE Trans. Syst., Man, Cybern.*, vol. SMC-3, no. 6, pp. 610–621, Nov. 1973, doi: 10.1109/TSMC.1973.4309314.

[15] L. Zhang, H. Liu, X. Gu, H. Guo, J. Chen, and G. Liu, "Sea ice classification using TerraSAR-X ScanSAR data with removal of scalloping and interscan banding," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 2, pp. 589–598, Feb. 2019, doi: 10.1109/JSTARS.2018.2889798.

[16] D. Murashkin, G. Spreen, M. Huntemann, and W. Dierking, "Method for detection of leads from Sentinel-1 SAR images," *Ann. Glaciol.*, vol. 59, no. 76pt2, pp. 124–136, 2018, doi: 10.1017/AOG.2018.6.

[17] D. Murashkin and G. Spreen, "Sea ice leads detected from sentinel-1 SAR images," in *Proc. Int. Geosci. Remote Sens. Symp.*, 2019, pp. 174–177, doi: 10.1109/IGARSS.2019.8898043.

[18] W. Aldenhoff, C. Heuzé, and L. E. B. Eriksson, "Comparison of ice/water classification in Fram Strait from C- and L-band SAR imagery," *Ann. Glaciol.*, vol. 59, no. 76pt2, pp. 112–123, 2018, doi: 10.1017/AOG.2018.7.

[19] N. Zakhvatkina, V. Smirnov, and I. Bychkova, "Satellite SAR data-based sea ice classification: An overview," *Geosciences*, vol. 9, no. 4, pp. 3–5, 2019, doi: 10.3390/geosciences9040152.

[20] J. Park, A. Korosov, M. Babiker, and H. Kim, "Automated sea ice classification using Sentinel-1 imagery," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Aug. 2019, pp. 2–5, doi: 10.1109/IGARSS.2019.8898731.

[21] X. Chen, Y. Kong, X.-B. Zhao, W.-J. Liu, and Y. Li, "SAR sea ice classification which is based on wavelet character," in *Proc. Int. Symp. Optoelectron. Technol. Appl. Image Process. Pattern Recognit.*, Nov. 2014, Art. no. 93011, doi: 10.1117/12.2072080.

[22] X. Xiang Zhu *et al.*, "Deep learning in remote sensing: A review," 2017, *arXiv:1710.03959*.

[23] H. Boulze, A. Korosov, and J. Brajard, "Classification of sea ice types in Sentinel-1 SAR data using convolutional neural networks," *Remote Sens.*, vol. 12, no. 13, pp. 1–20, 2020, doi: 10.3390/rs12132165.

[24] R. Ressel, A. Frost, and S. Lehner, "A neural network-based classification for sea ice types on X-band SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 7, pp. 3672–3680, Jul. 2015.

[25] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, vol. 9351, 2015, pp. 234–241, doi: 10.1007/978-3-319-24574-4-28.

[26] B. Cui, X. Chen, and Y. Lu, "Semantic segmentation of remote sensing images using transfer learning and deep convolutional neural network with dense connection," *IEEE Access*, vol. 8, pp. 116744–116755, 2020, doi: 10.1109/ACCESS.2020.3003914.

[27] S. Sun, L. Mu, L. Wang, and P. Liu, "L-UNet: An LSTM network for remote sensing image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2020, Art. no. 8004505, doi: 10.1109/LGRS.2020.3041530.

[28] L. C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder-decoder with atrous separable convolution for semantic image segmentation," in *Computer Vision—ECCV 2018* (Lecture Notes in Computer Science), vol. 11211, V. Ferrari, M. Hebert, C. Sminchisescu, and Y. Weiss, Eds. Cham, Switzerland: Springer, 2018, doi: 10.1007/978-3-030-01234-2_49.

[29] H. Fan, Q. Wei, D. Q. Shu, Y. Li, L. Zhang, and C. D. Yang, "An improved Deeplab based model for extracting cultivated land information from high definition remote sensing images," in *Proc. IEEE Int. Conf. Signal, Inf. Data Process. (ICSIDP)*, Dec. 2019, Art. no. 41471299, doi: 10.1109/ICSIDP47821.2019.9173010.

[30] Y. Lin, D. Xu, N. Wang, Z. Shi, and Q. Chen, "Road extraction from very-high-resolution remote sensing images via a nested SE-Deeplab model," *Remote Sens.*, vol. 12, no. 18, pp. 1–20, 2020, doi: 10.3390/RS12182985.

[31] H. Hersbach *et al.*, "ERA5 hourly data on single levels from 1979 to present," Copernicus Climate Change Service, Climate Data Store, 2018, doi: 10.24381/cds.adbb2d47.

[32] W. Dierking, "Mapping of different sea ice regimes using images from Sentinel-1 and ALOS synthetic aperture radar," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 3, pp. 1045–1058, Mar. 2010, doi: 10.1109/TGRS.2009.2031806.

[33] W. Dierking, "Sea ice monitoring by synthetic aperture radar," *Oceanography*, vol. 26, no. 2, pp. 100–111, 2013.

[34] S. K. Katiyar and P. V. Arun, "A review over the applicability of image entropy in analyses of remote sensing datasets," 2014, *arXiv:1405.6133*.

[35] B. Han and Y. Wu, "A novel active contour model based on modified symmetric cross entropy for remote sensing river image segmentation," *Pattern Recognit.*, vol. 67, pp. 396–409, Jul. 2017, doi: 10.1016/j.patcog.2017.02.022.

[36] D. P. Kingma and J. L. Ba, "Adam: A method for stochastic optimization," in *Proc. 3rd Int. Conf. Learn. Represent. Conf. (ICLR)*, 2015, pp. 1–15.

[37] S. C. Yurtkulu, Y. H. Sahin, and B. Bili, "Semantic segmentation with extended DeepLabv3 architecture," in *Proc. 27th Signal Process. Commun. Appl. Conf.*, 2019, pp. 10–13, doi: 10.1109/SIU.2019.8806244.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:58:59 UTC from IEEE Xplore. Restrictions apply.

[38] L. E. B. Eriksson *et al.*, "Evaluation of new spaceborne SAR sensors for sea-ice monitoring in the Baltic Sea," *Can. J. Remote Sens.*, vol. 36, no. 1, pp. S56–S73, Jan. 2010, doi: 10.5589/m10-020.

[39] A. M. Johansson, C. Brekke, G. Spreen, and J. A. King, "X-, C-, and L-band SAR signatures of newly formed sea ice in Arctic leads during winter and spring," *Remote Sens. Environ.*, vol. 204, pp. 162–180, Jan. 2018, doi: 10.1016/j.rse.2017.10.032.

Zeyu Liang received the M.S. degree from the School of Resources and Environmental Engineering, Anhui University, Hefei, China, in 2019. He is currently pursuing the Ph.D. degree in cartography and geographic information engineering with the Chinese Antarctic Center of Surveying, Wuhan University, Wuhan, China.

His research interests include sea ice remote sensing and deep learning.

Xiaoping Pang received the Ph.D. degree in solid geophysics from the School of Geodesy and Geomatics, Wuhan University, Wuhan, China, in 2007.

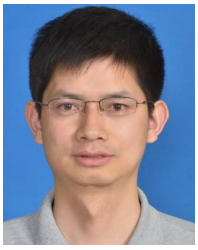
She is currently a Professor with the Chinese Antarctic Center of Surveying and Mapping, Wuhan University, and a member of the International Cartographic Association's Professional Committee on Data Sharing and Intellectual Property Protection. Her current main research interests include remote-sensing cartography and applications, geographic information visualization, and ice/snow environments dynamic simulation.

Qing Ji received the M.S. degree from Fujian Normal University, Fuzhou, China, in 2010, and the Ph.D. degree from Wuhan University, Wuhan, China, in 2015.

He is currently an Associate Professor with the Chinese Antarctic Center of Surveying and Mapping, Wuhan University. His research interests include data processing of satellite altimeter and sea ice remote sensing.

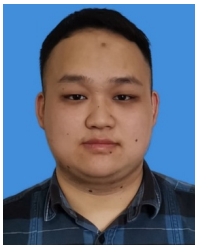
Xi Zhao received the Ph.D. degree in Earth observation science from the Faculty of Geo-Information Science and Earth Observation (ITC), University of Twente, Enschede, The Netherlands, in 2012.

From 2012 to 2021, she was an Associate Professor with the Chinese Antarctic Center of Surveying and Mapping, Wuhan University, Wuhan, China. Since September 2021, she has been an Associate Professor with the School of Geospatial Engineering and Science, Sun Yat-sen University, Zhuhai, China. Her research interests are sea ice remote sensing and uncertainty of spatial data.



Guoyuan Li received the B.Sc. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2007, the M.Sc. degree in photogrammetry and remote sensing from the Graduate School of Chinese Academy of Sciences, Beijing, China, in 2010, and the Ph.D. degree in cartography and geographic information engineering from Wuhan University, in 2017.

He is currently an Associate Researcher with the Land Satellite Remote Sensing Application Center, Ministry of Natural Resources, Beijing. His research interests include satellite laser altimetry and spaceborne photogrammetry and mapping.



Yizhuo Chen received the bachelor's degree from the School of Resource and Environmental Sciences, Wuhan University, Wuhan, China, in 2018, where he is currently pursuing the Ph.D. degree in cartography and geographic information engineering with the Chinese Antarctic Center of Surveying.

His main research direction is sea ice remote sensing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:58:59 UTC from IEEE Xplore. Restrictions apply.